

BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA

Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA
FANIDAN
TEST TOPSHIRIQLARI KITOB

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **TypoThesis jamoasi**

Telegram kanal: **@Milliymath_sertifikat**

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

– yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.

– ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'inishda yozing.

– moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!
1.03.2026. 1-smena.

1. Quyidagi sonni 7 ga bo'lgandagi qoldiqni toping:

$$\underbrace{77 \dots 79}_{2025 \text{ ta}} + \underbrace{77 \dots 78}_{2025 \text{ ta}} + \dots + \underbrace{77 \dots 71}_{2025 \text{ ta}} .$$

- A) 1
- B) 3
- C) 5
- D) 4

2. Hisoblang (Bu yerda, $[a]$ - a sonning butun qismi, $\{a\}$ - a sonining kasr qismi.):

$$\left\{ 3[\lg 5] - 5 \left\{ \lg \frac{1}{10} \right\} + 7[\ln 3] \right\} .$$

- A) 0
- B) 2
- C) 1
- D) 3

3. 40 m uzunlikdagi arqondan eng katta yuzaga ega bo'lgan shakl yasaldi. Bu shakl yuzini toping.

- A) 400
- B) $\frac{200}{\pi}$
- C) $\frac{400}{\pi}$
- D) 200

4. Dengiz suvining sho'rli 5 %. 30 litr suvga qancha toza suv qo'shilsa uning foizi 2 foizga tushadi.

- A) 42 l
- B) 45 l
- C) 50 l
- D) 54 l

5. Hisoblang:

$$\left(\left(3 \cdot 24^{\frac{1}{2}} \right)^{\frac{1}{3}} + 2^{\frac{1}{3}} \cdot 54^{\frac{1}{6}} \right)^2 .$$

- A) 12
- B) 18
- C) 20
- D) 24

6. Hisoblang:

$$\sqrt[6]{5^7 \cdot \sqrt[7]{5} \cdot \sqrt[7]{5^{-2}}}.$$

- A) $\sqrt[20]{5^{19}}$
- B) $\sqrt[21]{5^{20}}$
- C) $\sqrt[21]{5^{19}}$
- D) $\sqrt[21]{5^9}$

7. Hisoblang:

$$\frac{5}{\sqrt[3]{32} + 2} + \frac{5}{\sqrt[3]{128} - \sqrt[3]{32} + 2} - \sqrt[3]{2}.$$

- A) 2
- B) 0
- C) $\sqrt[3]{2} + 1$
- D) 1

8. a_n arifmetik progressiyada $a_2 = 2a_1$ va $a_{10} = 20$ bo'lsa, $S_6 - S_5$ ni toping.

- A) 16
- B) 10
- C) 12
- D) 14

9. $b_1 + b_2 + b_3 = 217$ va $b_1 = a_2$, $b_2 = a_9$, $b_3 = a_{44}$ hamda $a_1 + a_2 + \dots + a_n = 820$ bo'lsa n ni toping. Bu yerda a_n arifmetik progressiya, b_n geometrik progressiya.

- A) 20
- B) 19
- C) 18
- D) 21

10. Soddalashtiring:

$$\left(1 - \frac{x^2 + y^2 - 4}{2xy}\right) : \frac{x - y + 2}{x - y - 2} : \frac{2 - x + y}{2xy}.$$

- A) $x + y + 2$
- B) $x - y + 2$
- C) $y - x - 2$
- D) $x - y - 2$

11. $(a + b + 2025)^2 = 4(a + 1012)(b + 1013)$ bo'lsa, $a - b$ ni toping.

- A) -2025
- B) 1
- C) -2026
- D) 0

12. Hisoblang:

$$\frac{\sin 40^\circ - \cos 40^\circ}{\sqrt{2} \cos 85^\circ} .$$

- A) 1
- B) -1
- C) $\cos^2 85$
- D) 2

13. Tenglamani yeching:

$$2025^{x^2-4} = 2026^{x^2-4} .$$

- A) ± 1
- B) 2
- C) ± 2
- D) -2

14. Tengsizlikni nechta butun son qanoatlantiradi:

$$\frac{(x-1)^2 \sqrt{x+6}}{x^2 - 12x + 35} \leq 0 .$$

- A) 2 ta
- B) 3 ta
- C) Haqiqiy ildizga ega emas
- D) 1 ta

15. Tenglamani yeching:

$$\sqrt{x-2} - \sqrt{6-x} = 2 .$$

- A) 4
- B) 5
- C) 6
- D) 8

16. Tengsizlikning butun yechimlari soni toping:

$$\frac{2x^2 - 14x + 6}{x^2 - 4x + 3} - \frac{3x - 8}{x - 3} \geq 0 .$$

- A) $[-2; -1] \cup (1; 3)$
- B) $(-2; -1) \cup (1; 3)$
- C) $[-2; -1)$
- D) $[-2; -1]$

17. Tenglamalar sistemasining yechimlari uchun $x + y$ ni toping:

$$\begin{cases} \log_3(3 \cdot 5^x + 2 \cdot 3^y) = 4 \\ 5^{x+1} - 4 \cdot 3^y = 113. \end{cases}$$

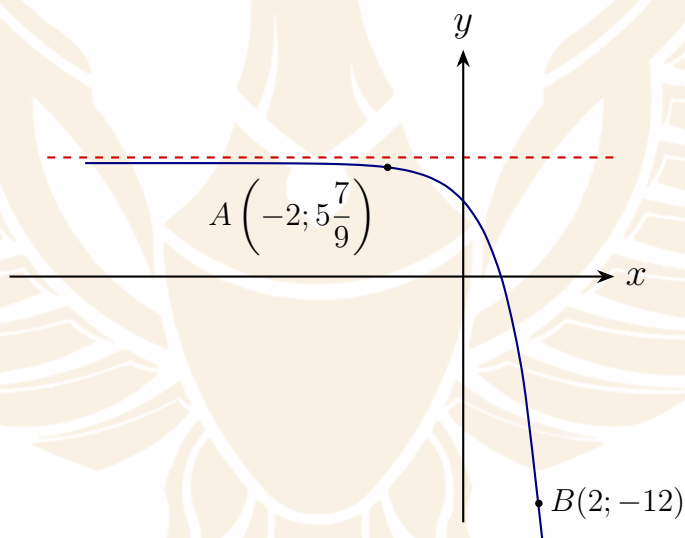
- A) 3
- B) 4
- C) 2
- D) 5

18. Tengsizlikni yeching:

$$\sqrt{3} \cos 2x - 6 \sin x - 3\sqrt{3} < 0.$$

- A) $\frac{\pi}{4} + 2\pi k, k \in \mathbb{Z}$
- B) $\frac{\pi}{2} + \pi k, k \in \mathbb{Z}$
- C) $x \in \emptyset$
- D) $x \in \mathbb{R}$

19. $f(x) = a \cdot 3^x + b$ funksiya $A\left(-2; 5\frac{7}{9}\right)$ va $B(2; -12)$ nuqtalardan o'tadi. Funksiyaga teskari funksiya aniqlanish sohasini toping.



- A) $(6; \infty)$
- B) $(-6; \infty)$
- C) $(-\infty; 6)$
- D) $(-\infty; -6)$

20. $y = f(x)$ funksiya juft funksiya va $y = g(x)$ funksiya esa toq funksiya bo'lsin. Quyidagilardan nechitasi toq funksiya bo'ladi:

1. $y = f(x) \cdot g(x);$
2. $y = \frac{f(x)}{g(x)};$
3. $y = f(x) + g(x).$

- A) 0
- B) 1
- C) 2
- D) 3

21. $f(x) = (x - 8)e^{x-7}$ funksiyaning $[5; 9]$ oraliqdagi eng kichik qiymatini toping.

- A) e^3
- B) -1
- C) e^{-2}
- D) e^2

22. $y = -x + 1$ va $x^2 - 5x + 6$ funksiyalarning grafiklari orasidagi eng qisqa masofani toping.

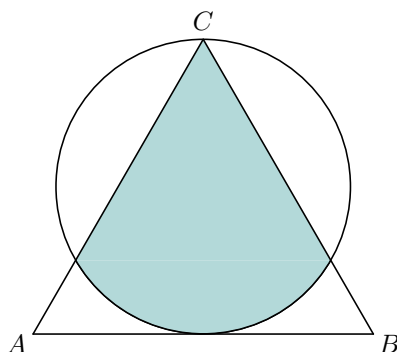
- A) $\frac{\sqrt{3}}{2}$
- B) 0
- C) $\frac{\sqrt{2}}{2}$
- D) $\frac{1}{2}$

23. Aniqmas integralni toping:

$$\int \frac{dx}{x^4 + 1}.$$

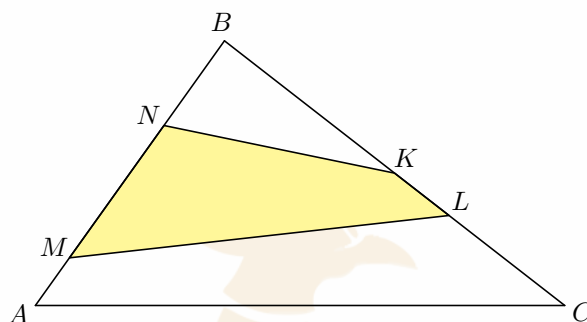
- A) $\frac{1}{2\sqrt{2}} \ln \left| \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} \right| - \frac{1}{\sqrt{2}} \arctg \left(\frac{\sqrt{2}x}{1 - x^2} \right) + C$
- B) $\frac{1}{4\sqrt{2}} \ln \left| \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} \right| - \frac{1}{\sqrt{2}} \arctg \left(\frac{\sqrt{2}x}{1 - x^2} \right) + C$
- C) $\frac{1}{2\sqrt{2}} \ln \left| \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} \right| + \frac{1}{\sqrt{2}} \arctg \left(\frac{\sqrt{2}x}{1 - x^2} \right) + C$
- D) $\frac{1}{4\sqrt{2}} \ln \left| \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} \right| + \frac{1}{\sqrt{2}} \arctg \left(\frac{\sqrt{2}x}{1 - x^2} \right) + C$

24. ABC muntazam uchburchakning tomoni 12 ga teng bo'lsa, bo'yalgan soha yuzini toping.



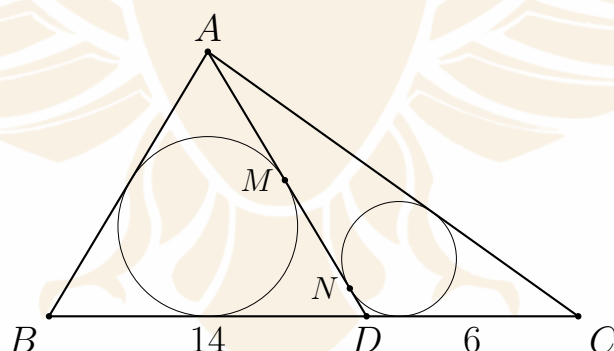
- A) $\frac{27\sqrt{3}}{2} + 9\pi$
 B) $\frac{9\sqrt{3}}{2} + 9\pi$
 C) $\frac{18\sqrt{3}}{2} + 9\pi$
 D) $\frac{27\sqrt{3}}{2} - 9\pi$

25. ABC uchburchakning yuzi 108 ga teng. $AM : MN : NB = 1 : 3 : 2$, $BK : KL : LC = 3 : 1 : 2$ bo'lsa, bo'yalgan soha yuzini toping.



- A) 84
 B) 42
 C) 36
 D) 21

26. ABC teng yonli uchburchakning BC asosiga AD kesma tushirilgan va ABD , ADC uchburchaklarga ichki aylanalar chizilgan. Agar, $BD = 14$, $DC = 6$ bo'lsa, MN kesma uzunligini toping.



- A) 4
 B) 6
 C) 2
 D) 5

27. Trapetsiyaning asoslari 16 va 44, yon tomonlari 17 va 25. Trapetsiya yuzini toping.

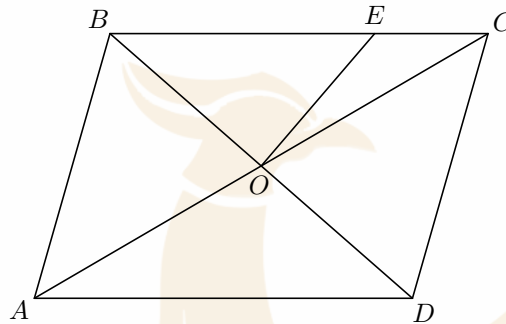
- A) 420
 B) 450

- C) 360
D) 520

28. Muntazam ko'pburchakning diogonallari soni 35 ta bo'lsa, bir uchidan chiqqa eng katta va eng kichik diogonallari orasidagi burchak.

- A) 72°
B) 54°
C) 36°
D) 45°

29. $ABCD$ parallelogramda O nuqta dioganallar kesishish nuqtasi. $BE : EC = 3 : 1$ ga bo'lsa, $\overrightarrow{OE} = a \cdot \overrightarrow{AB} + b \cdot \overrightarrow{AD}$ bo'lsa, $a + b$ ni toping.



- A) 1
B) $\frac{1}{2}$
C) $\frac{3}{4}$
D) $\frac{3}{5}$

30. Radiusi 4 va 12 ga teng bo'lgan kesik konusni yon sirti yuzi umumiy yuzasidan 2 marta kichik bo'lsa konusni hajmini toping.

- A) 512π
B) 416π
C) 432π
D) 448π

31. 25 nafar o'quvchi bor. 9 ta o'quvchi biologiya to'garagiga boradi. 8 ta o'quvchi kimyo to'garagiga boradi. 10 ta o'quvchi hech qaysisiga bormaydi. Nechta o'quvchi 2 ta to'garakkayam boradi.

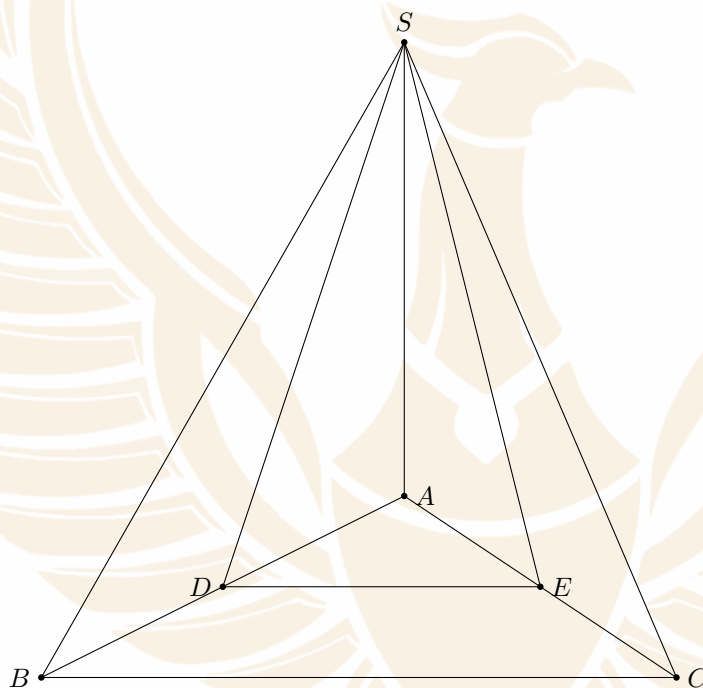
- A) 1
B) 2
C) 3
D) 4

32. “MATEMATIKA” soʻzidagi harflar qogʻozga alohida yozib chiqilgan. Harflar ichidan tasodifiy tanlanganda M harfi chiqish ehtimolini toping.

- A) $\frac{1}{10}$
 B) $\frac{1}{5}$
 C) $\frac{1}{4}$
 D) $\frac{2}{5}$

Topshiriqlar (33–35) va javob variant (A–F) larni oʻzaro moslashtiring.

$SABC$ muntazam uchburchakli piramidaning tomoni 2 ga teng. D va E nuqtalar mos ravishda AB va AC tomoni oʻrtalari. SDE kesim asos tekisligi bilan 45° hosil qiladi.



Javoblar.

- A) $\frac{\sqrt{6}}{12}$
 B) $\frac{\sqrt{30}}{4}$
 C) $\frac{\sqrt{15}}{6}$
 D) $\frac{\sqrt{15}}{4}$
 E) $\frac{1}{8}$
 F) $\frac{1}{6}$

33. SE ni toping.

34. S_{SDE} ni toping.

35. V_{SDECB} ni toping.

36. Tenglamani yeching:

$$x^4 - 2\sqrt{3}x^2 - x + 3 - \sqrt{3} = 0 .$$

a) Tenglama nechta turli musbat ildizga ega?

Javob: _____

b) Tenglama nechta turli haqiqiy ildizga ega?

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

37. Tenglamani yeching:

$$\frac{\operatorname{tg} 2x \cdot \operatorname{tg} x}{\operatorname{tg} 2x - \operatorname{tg} x} = 2 \sin(45^\circ + x) \cdot \sin(45^\circ - x) .$$

a) Eng katta manfiy ildizini toping.

Javob: _____

b) Tenglamaning $[0; \pi]$ oralig'ida nechta ildizga ega.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

38. $f(x) = x^3 + ax^2 + bx + c$ funksiya uchun $(x+1)f(x-1) + (x-1)f(x+1) = 2xf(x)$ tenglik o'rinli.

a) $f(1) - f(-1)$ ni toping.

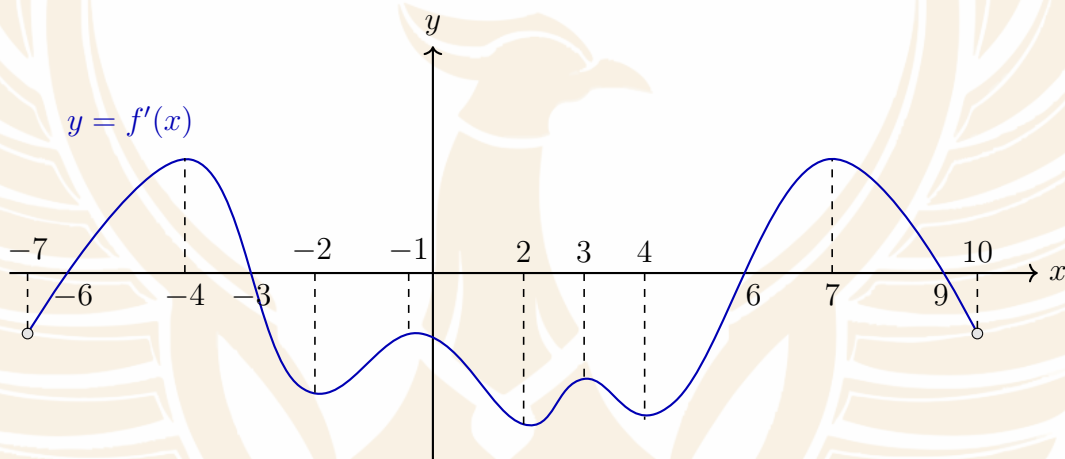
Javob: _____

b) $f(2) - f(-2)$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

39. Quyida $(-7; 10)$ oralig'ida aniqlangan $f(x)$ funksiyaning hosilasi grafigi tasvirlangan.



a) $f(x)$ funksiyaning ekstremum nuqtalari nechta?

Javob: _____

b) $\frac{f'(x)}{f''(x)} < 0$ tengsizlikni $(-7; 10)$ oralig'idagi nechta butun son qanoatlantiradi.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

40. Dekart koordinatalar sistemasida $y = x^3$ egri chiziqliq, $x = 0$ va $y = -1$ to'g'ri chiziqlar tasvirlangan.

a) Bu funksiyalar bilan chegaralangan soha yuzini toping.

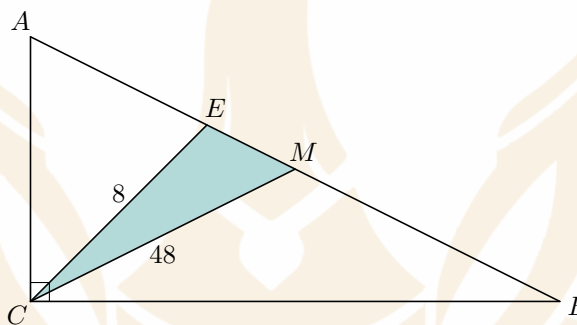
Javob: _____

b) Bu funksiyalar bilan chegaralangan soha yuzini Oy o'qi atrofida 360° aylantirishdan hosil bo'lgan jism hajmini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

41. To'g'ri burchakli uchburchakning to'g'ri burchagi uchidan chiqqan medianasi va bissektrisasi mos ravishda 48 va 8.



a) Uchburchak yuzini toping.

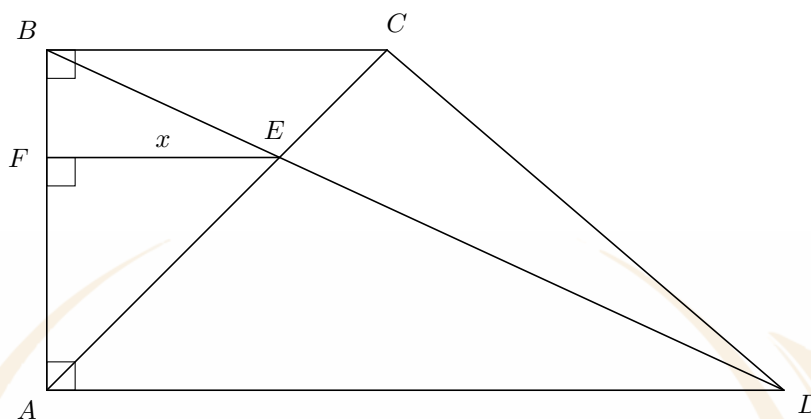
Javob: _____

b) Bo'yalgan soha yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

42. Asoslari 3 va 7, kichik yon tomoni 5 ga teng bo'lgan to'g'ri burchakli trapetsiya berilgan.



a) Dioganallari kesishish nuqtasidan kichik yon tomongacha bo'lgan x masofani toping.

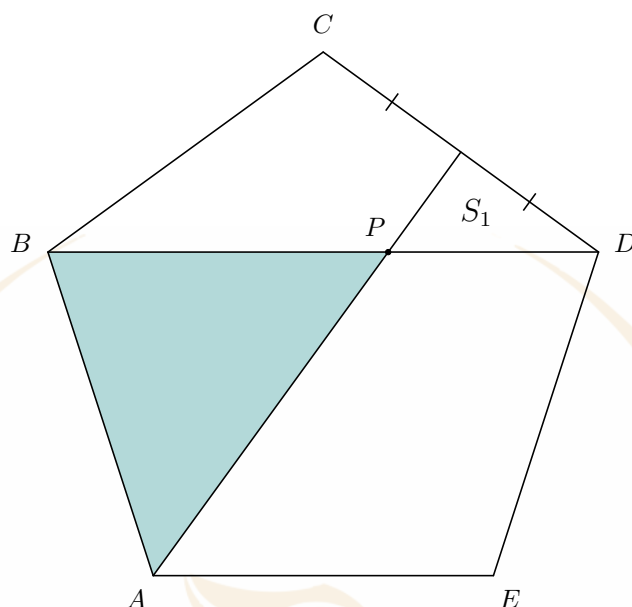
Javob: _____

b) S_{ECD} ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

43. $ABCDE$ muntazam beshburchakning BD diagonali o'tkazilgan. CD ning o'rtasi va A uchi tutashtirildi.



a) $PD = \sqrt{5} - 1$ bo'lsa, BP ni toping.

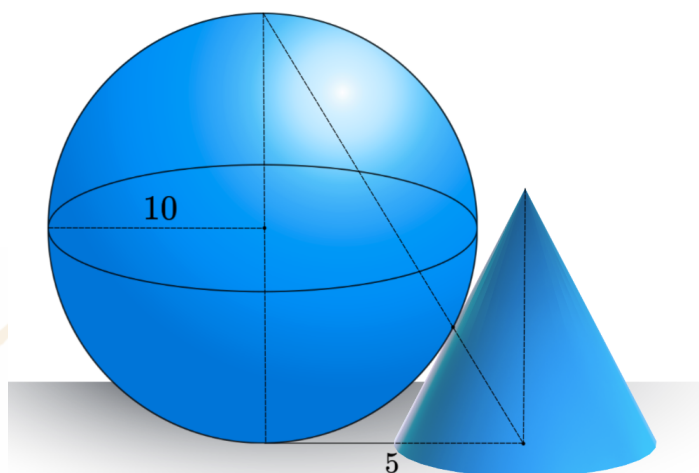
Javob: _____

b) $S_1 = 3 - \sqrt{5}$ bo'lsa, bo'yalgan soha yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiqa ko'chirib yozing.

44. Rasmda konus va unga tayanib turgan radiusi 10 ga teng bolgan sfera tasvirlangan. (Bunda $\pi = 3$) deb oling.)



a) Konusning balandligini toping.

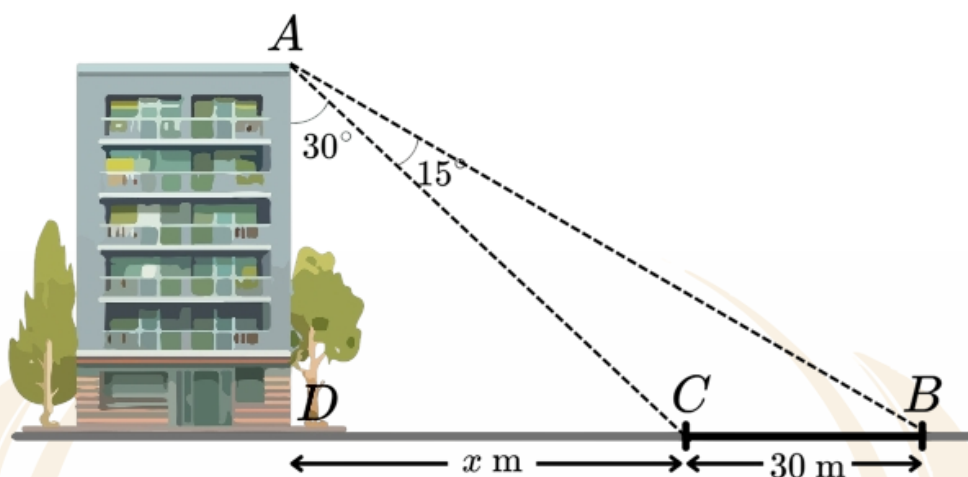
Javob: _____

b) Konusning hajmini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

45. Rasmda bino va uning yuqori nuqtasidan yerdagi orasidagi masofa 30 m bo'lgan ikki nuqtaning ko'rinish burchaklari berilgan.



a) Binoning balandligini toping.

Javob: _____

b) AB ning uzunligini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.